

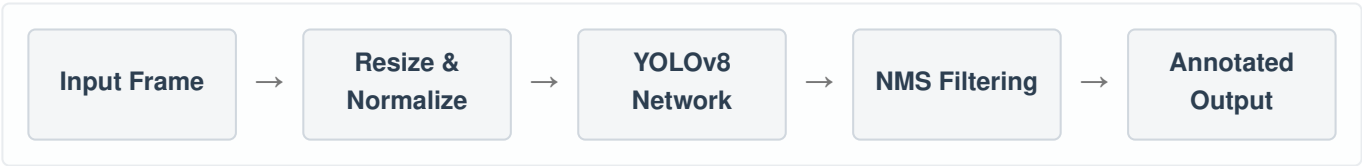
Object Detection with Python

HACKATHON PROJECT SUBMISSION REPORT

1. Algorithm Overview

- **Model/Technique:** YOLOv8 (You Only Look Once) real-time object detector implemented in Python.
- **Logic/Method:** Single-pass anchor-free feature extraction using CloU loss for box regression and Non-Maximum Suppression (NMS) to clear overlapping boxes.
- **Step-by-Step Procedure:**
 - Load pre-trained weights (yolov8n.pt) using the ultralytics package.
 - Preprocess input frames (resize to 640 × 640 pixels).
 - Pass image through the Python network for inference.
 - Apply NMS filter to final bounding boxes.
 - Render labeled outputs via OpenCV (cv2).

Process Flow:



2. Dataset Overview

- **Source of Data:** MS COCO / Roboflow hosted dataset.
- **Number of Records/Samples:** 10,000 total image samples.
- **Data Format or Type:** .jpg images paired with normalized .txt YOLO annotation coordinates.
- **Features or Attributes:** Class ID, X_Center, Y_Center, Width, Height (all bounding box parameters normalized between 0.0 and 1.0).

Training and Testing Data Split:

Split	Percentage	Sample Count
Train	70%	7,000 images
Val	20%	2,000 images
Test	10%	1,000 images